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Author(s): Emily K. Smith-Mckenna, Lynn M. Resler, Diana F. Tomback, Huaiye Zhang & George P. Malanson

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Topographic influences on the distribution of white pine blister rust in *Pinus albicaulis* treeline communities¹

Emily K. SMITH-MCKENNA² & Lynn M. RESLER, Department of Geography, Virginia Tech, 115 Major Williams Hall (0115), Blacksburg, Virginia 24061, USA, e-mail: emksmith@vt.edu

Diana F. TOMBACK, Department of Integrative Biology, CB 171, University of Colorado Denver, PO Box 173364, Denver, Colorado 80217, USA.

Huaiye ZHANG, Department of Statistics, Virginia Tech, 406-A Hutcheson Hall, Blacksburg, Virginia 24061, USA.

George P. MALANSON, Department of Geography, University of Iowa, 316 Jessup Hall, Iowa City, Iowa 52242, USA.

Abstract: The exotic disease white pine blister rust (caused by *Cronartium ribicola*) damages and kills whitebark pine (*Pinus albicaulis*), even in the extreme environments of alpine treeline communities. We surveyed *P. albicaulis* trees and tree islands for blister rust in 2 distinct alpine treeline communities in Montana, USA, and examined meso- and microtopographic factors potentially related to the climatic requirements for blister rust infection. For each of 60 sampling plots, we created high-resolution digital elevation models, derived microtopography variables, and compared these and distance to water feature variables with blister rust occurrence and intensity (number of cankers per infected tree) for every sampled *P. albicaulis* tree. Infection rates were 19% (of 328 sampled trees) and 24% (of 585 sampled trees) at the 2 sites. Tree island *P. albicaulis* had higher infection percentages than solitary trees. Using Bayesian analysis and a zero-inflated Poisson regression model, we determined that solar radiation and moisture-related variables correlated with both presence and number of blister rust cankers on *P. albicaulis*. Site factors that influence moisture, such as local topography, hydrology, and climate, differed between the 2 treeline study areas, which may account for the model variability.

Keywords: *Cronartium ribicola*, digital elevation models (DEM), Rocky Mountains, treeline, whitebark pine, zero-inflated Poisson regression.

Résumé: La rouille vésiculeuse du pin blanc, une maladie exotique causée par *Cronartium ribicola*, endommage et tue le pin à écorce blanche (*Pinus albicaulis*) même dans des communautés d'environnements extrêmes comme à la limite alpine des arbres. Nous avons échantillonné des individus et des îlots de *P. albicaulis* pour la rouille vésiculeuse dans 2 communautés distinctes de la limite alpine des arbres au Montana (États-Unis). Nous avons examiné les facteurs méso et microtopographiques potentiellement liés aux exigences climatiques pour une infection par la rouille vésiculeuse. Pour chacune des 60 parcelles d'échantillonnage, nous avons créé des modèles numériques d'élévation de haute résolution et dérivé des variables microtopographiques, nous avons ensuite comparé celles-ci et des variables liées à la distance à l'eau avec la présence et l'intensité de la rouille vésiculeuse (nombre de chancres par arbre infecté) pour chaque individu de *P. albicaulis* échantillonné. Les taux d'infection des 2 sites étaient de 19 % (328 arbres échantillonnés) et de 24 % (585 arbres échantillonnés). Les îlots de *P. albicaulis* avaient des pourcentages d'infection plus élevés que ceux d'arbres isolés. En utilisant l'analyse bayésienne et un modèle de régression de Poisson avec surreprésentation de zéros, nous avons déterminé que des variables liées au rayonnement solaire et à l'humidité étaient corrélées avec la présence et le nombre de chancres de rouille vésiculeuse sur *P. albicaulis*. Les facteurs qui influencent l'humidité, comme la topographie locale, l'hydrologie et le climat, différaient entre les 2 sites d'étude ce qui peut avoir contribué à la variabilité du modèle.

Mots-clés: *Cronartium ribicola*, limite des arbres, modèles numériques d'élévation, montagnes Rocheuses, pin à écorce blanche, régression de Poisson avec surreprésentation de zéros.

Nomenclature: Flora of North America Editorial Committee, 1993; Integrated Taxonomic Information System, online.

Introduction

Landscape pathology integrates concepts from disease ecology and landscape ecology and emphasizes pattern–process relationships between vegetation and disease manifestation at a landscape scale (e.g., Jules *et al.*, 2002; Holdenrieder *et al.*, 2004; Condeso & Meentemeyer, 2007; Meentemeyer *et al.*, 2011). Variation in meso- (e.g., spanning across watersheds) and

micro- (e.g., local, small depressions) topographic landscape features may influence the spatial distribution of host and disease reservoir species, pathogen spread, and disease manifestation (e.g., Rizzo *et al.*, 2002; White, Brown & Host, 2002; Wilson, Lewis & Aberton, 2003; Holdenrieder *et al.*, 2004; Kearns & Jacobi, 2007; Larson, 2011; Smith *et al.*, 2011). Since specific abiotic conditions resulting from topographic heterogeneity can affect disease incidence (e.g., Smith & Hoffman, 2001; White, Brown & Host, 2002; Kearns & Jacobi, 2007; Smith *et al.*, 2011), a landscape pathology approach for pathogens with complex life cycles may relate how disease manifestation and

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²Author for correspondence.

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affected vegetation dynamics are expressed in mountainous regions, especially under scenarios of global environmental change (e.g., Shafer, Bartlein & Whitlock, 2005).

The introduced fungal pathogen *Cronartium ribicola*, which causes white pine blister rust (here referred to as blister rust) in five-needle white pines (genus *Pinus*, subgenus *Strobus*) is a problematic plant pathogen with a wide distribution and complex life cycle (Hoff, Bingham & McDonald, 1980). The life cycle comprises 5 stages of spore production and windborne transmission between the white pine host and alternate hosts, typically *Ribes* spp. but also *Pedicularis racemosa* and *Castilleja miniata* (Spaulding & Gravatt, 1917; McDonald *et al.*, 2006). Spore production and transmission between hosts require environments with moderate temperatures, humidity, and wind conditions (McDonald & Hoff, 2001). In the extreme environments of the alpine treeline ecotone (ATE), a transition zone between closed contiguous upright trees and treeless tundra (Holtmeier, 2009), blister rust now occurs in *Pinus albicaulis* (whitebark pine) treeline communities in the northern Rocky Mountains east of the Continental Divide (Resler & Tomback, 2008; Smith *et al.*, 2008; Smith *et al.*, 2011). These environments are exposed to conditions previously thought too dry and cold for blister rust spore production (e.g., Van Arsdel, Riker & Patton, 1956; Campbell & Antos, 2000).

Inadvertently introduced to western North America in the early 1900s, *Cronartium ribicola* now occurs nearly range-wide in *P. albicaulis*, a highly susceptible species, and has been responsible for precipitous population declines in the northern Rocky Mountains (Hoff, Bingham & McDonald, 1980; Tomback & Achuff, 2010). Blister rust cankers girdle branches, ending seed production and reducing photosynthetic biomass; cankers in main stems kill trees (McDonald & Hoff, 2001; Geils, Hummer & Hunt, 2010). *Pinus albicaulis* functions as a foundation and keystone species by fostering community development and biodiversity across its range in the US and Canada (e.g., Tomback, Arno & Keane, 2001; Ellison *et al.*, 2005). The species depends primarily on the Clark's nutcracker (*Nucifraga columbiana*) for dispersal of its large, wingless seeds (e.g., Tomback, 1978; 1982; 2001). Loss of seed production has resulted in the decline of *P. albicaulis* seed dispersal by nutcrackers as well as a decline in *P. albicaulis* regeneration in the northern US Rocky Mountains (McKinney, Fiedler & Tomback, 2009; Barringer *et al.*, 2012).

Blister rust spore production, transmission to alternate hosts, and infection of *P. albicaulis* require specific temperature, humidity, and wind conditions (Mielke, 1943; Van Arsdel, Riker & Patton, 1956; Smith & Hoffman, 2000; McDonald & Hoff, 2001; Smith & Hoffman, 2001; Frank *et al.*, 2008; Rochefort, 2008). Studies of *Ribes* and white pine hosts show the optimal conditions required for each spore development stage vary in temperature (ranging from 13 °C to 20 °C), moisture, transport distances (both local and long distance), and length of time (hours, days, or weeks) to complete the blister rust life cycle (Hirt, 1942; Van Arsdel, Riker & Patton, 1956; McDonald & Andrews, 1980). For example, the final stage and development

of fragile basidiospores, which infect white pine hosts, requires cool temperatures (less than 20 °C), 48 h of near 100% relative humidity (Van Arsdel, Riker & Patton, 1956), limited sun and frost exposure, and wind to transport basidiospores from *Ribes* spp. over short distances (less than 300 m) (McDonald & Hoff, 2001).

Topography influences a hierarchy of abiotic conditions by moderating meso- and microclimates over a range of spatial scales, especially at alpine treelines, where tree physiology is limited by adverse climatic conditions (e.g., Holtmeier, 2009). In mountain environments, topography affects environmental controls at local to regional spatial scales, including (but not limited to) local soil structure and moisture, freeze-thaw processes, snow accumulation, and local climate in sheltered *versus* exposed sites (Johnson & Billings, 1962; Oke, 1978). Alpine microtopography has been linked to species richness, microclimate conditions, and microsite variation, which can determine the survival or demise of seedlings (e.g., Germino, Smith & Resor, 2002; Butler *et al.*, 2007; Rose & Malanson, 2012). Microtopographic features such as depressions and boulders can shelter and promote growth of conifer seedlings (Resler, 2004) by reducing wind exposure and snow abrasion in severe ATE environments (Holtmeier, 2009). Concurrently, these sheltered areas may also provide higher humidities that favour fungal diseases (Oke, 1978) such as blister rust.

Abiotic conditions can be derived from topography through the use of digital elevation models (DEMs), which are effective for deriving surrogates of field-collected environmental variables such as slope and moisture (Brown, 1994) and solar radiation (Guisan, Theurillat & Kienast, 1998) in models of mountain landscapes (Del Barrio *et al.*, 1997; Bader & Ruijten, 2008). Environmental gradients resulting from complex topography have been associated with varying levels of blister rust infection severity (e.g., White, Brown & Host, 2002; Kearns & Jacobi, 2007; Smith *et al.*, 2011), the distribution of *Ribes* spp. (which vary in their susceptibility to *Cronartium ribicola* infection: Newcomb, 2003; Newcomb, Upper & Rouse, 2010), and ability of *Ribes* spp. to support production of pine-infecting basidiospores (Childs & Kimmey, 1938; Kearns *et al.*, 2008). For example, higher incidence of blister rust-infected *Pinus flexilis* (limber pine) and *P. albicaulis* have been associated with topographic depressions (Kearns & Jacobi, 2007; Smith *et al.*, 2011) and proximity to perennial streams, which may increase humidity and the occurrence of some riparian species of *Ribes* (Newcomb, 2003).

The decline of *P. albicaulis* as a result of blister rust may alter treeline community structure by disrupting the dynamics of tree island development at some Rocky Mountain treelines. In *P. albicaulis* treeline communities, krummholz (environmentally stunted) forms of *P. albicaulis* typically shelter conifers less tolerant of cold and windy conditions, resulting in the formation of tree island colonies, a notable treeline pattern (e.g., Habeck, 1969; Resler, 2004; Resler & Tomback, 2008; Resler & Fonstad, 2009). For example, near Logan Pass in Glacier National Park, Habeck (1969) predicted that high blister rust infection levels could alter the dynamics of meadow invasion by trees, given the facilitative relationship between

P. albicaulis and *Abies lasiocarpa* (subalpine fir) in that area. Resler and Tomback (2008) noted that blister rust is killing established *P. albicaulis* trees in the ATE before they facilitate the recruitment of other conifers, hastening the decline of *P. albicaulis* and the facilitation of tree islands, as well as killing *P. albicaulis* established within tree islands. Furthermore, subalpine *P. albicaulis* seed sources are diminished, resulting in fewer seeds available for dispersal by nutcrackers to the ATE (McKinney & Tomback, 2007; McKinney & Fiedler, 2010). The loss of *P. albicaulis* to blister rust at alpine treeline is likely to slow and/or diminish the predicted upslope advance of treeline as climate warms (Tomback & Resler, 2007). Treeline position in some regions may serve as a bioclimatic indicator of climate change (Körner & Paulsen, 2004), but interpretations of treeline position may be confounded in treeline ecosystems where *P. albicaulis* is the predominant conifer species, because blister rust-induced mortality of *P. albicaulis* may impact treelines that must respond to both a warming climate and a disease-altered ecosystem (E. Smith-McKenna *et al.*, unpubl. data).

The presence of blister rust in the dry and cold climates of *P. albicaulis* treelines in the northern Rocky Mountains east of the Continental Divide (Resler & Tomback, 2008; Smith *et al.*, 2008; Smith *et al.*, 2011) is perplexing since controlled experiments suggest that blister rust spore production requires moist and moderately cool environments to thrive (Van Arsdel, Riker & Patton, 1956). Previous research indicates that vegetation pattern and structure and topography influence the rate of blister rust infection in *P. albicaulis* treeline communities (Resler & Tomback, 2008; Smith *et al.*, 2011). These studies have been exploratory in nature and have primarily focused on characterizing general relationships between fine-scale topographic and environmental variables and blister rust infection rates, aggregated at the sampling plot level (*e.g.*, Smith *et al.* 2011). However, it is unknown how infection presence and intensity varies spatially from one *P. albicaulis* tree to another, and how this spatial variation is influenced by the underlying and surrounding topography of the tree. Given the topographic influences on abiotic conditions over a range of spatial scales at alpine treelines (Malanson *et al.*, 2007; Holtmeier, 2009), there is merit in investigating the link between topography and blister rust infection through a multiscale analysis that considers the impact at the tree level in addition to broader spatial scales of influence. Here, we investigate topographic factors (both meso- and micro-) that are potentially related to blister rust infection of *P. albicaulis* trees.

The purpose of this study is to examine the relationships between blister rust infection and ecologically significant topographic variables at the tree level for 2 alpine treeline study locations in the US Rocky Mountains. Our objectives were to 1) survey the presence and intensity of blister rust infection (if any) in 2 geographically different *P. albicaulis* treeline communities in Montana, USA, the first in northern Montana, near the pine's mid-latitudinal distribution, and the second in the Beartooth Plateau, near the southernmost region of the pine's Rocky Mountain range; 2) determine which meso- and microtopographic

factors affecting moisture and exposure conditions at a tree's site correlate with the presence and intensity of blister rust infection among the sampled *P. albicaulis* trees; and 3) assess any geographic differences in blister rust infection and potential influential factors between the 2 treelines.

Methods

STUDY AREAS

Working at treeline in the Northern Divide Ecosystem, specifically in Glacier National Park, we previously found that *P. albicaulis* was an important facilitator of tree island development (Resler & Tomback, 2008). Additional work in this region in 2008 determined significant relationships between topography and areas of blister rust infection at treeline at the plot level (Smith *et al.*, 2011). In 2010 we extended our work by increasing sampling intensity and geographic coverage, sampling in 2 treeline study areas along the Rocky Mountain Front of Montana. *Pinus albicaulis* is a major treeline component in both study areas. The 2 study areas differ in surrounding topography but have similar microtopography; that fact and their separation by 3° of latitude and 500 km distance enabled us to sample from *P. albicaulis* communities across a range of environmental conditions (Figure 1). The study areas are located near the mid- and southern Rocky Mountain range of *P. albicaulis* east of the Continental Divide in the Rocky Mountains of Montana, USA, in areas dominated by dry, continental climate regimes (Johnson & Billings, 1962; Finklin, 1986).

Our northern study area, Divide Mountain (about 48.6°N, 113.4°W), crosses the eastern boundary of Glacier National Park (GNP) and western boundary of the Blackfoot Indian Reservation along the eastern Rocky Mountain Front, within the Northern Divide Ecosystem. Here, *P. albicaulis* is a major component of krummholz forest communities, where we sampled at elevations ranging from 2097 to 2261 m, on the slopes of Divide and White Calf Mountains (Figure 1). To characterize summer climate conditions during the sampling season, we measured temperatures at the Divide Mountain study area using 2 Hobo® (Onset Computer Corporation, Bourne, MA, USA) weather stations (about 2 m tall) from July through September 2010. Summer temperatures in 2010 averaged 9.6 °C ($T_{\min} = -1.4$ °C, $T_{\max} = 22.2$ °C).

The southern study area lies just north of the Wyoming border on the northeastern edge of the Beartooth Plateau (part of the Greater Yellowstone Ecosystem [GYE]), in the Line Creek Research Natural Area (RNA), Custer National Forest (about 45.0°N, 109.4°W) (Figure 1). Mellmann-Brown (2005) found *P. albicaulis* to be a predominant conifer species in treeline communities within the Line Creek RNA. Here, we sampled treeline communities on a narrow, exposed, and wind-swept ridge bounded by steep slopes above Wyoming Creek, at elevations ranging from 2960 to 3029 m. Temperatures measured at the Line Creek RNA (July–September 2010) averaged 8.9 °C ($T_{\min} = -8.6$ °C, $T_{\max} = 22.1$ °C).

At both study areas, surface microtopographic features occurring throughout the landscape include topographic depressions, turf-banked terraces, patterned ground,

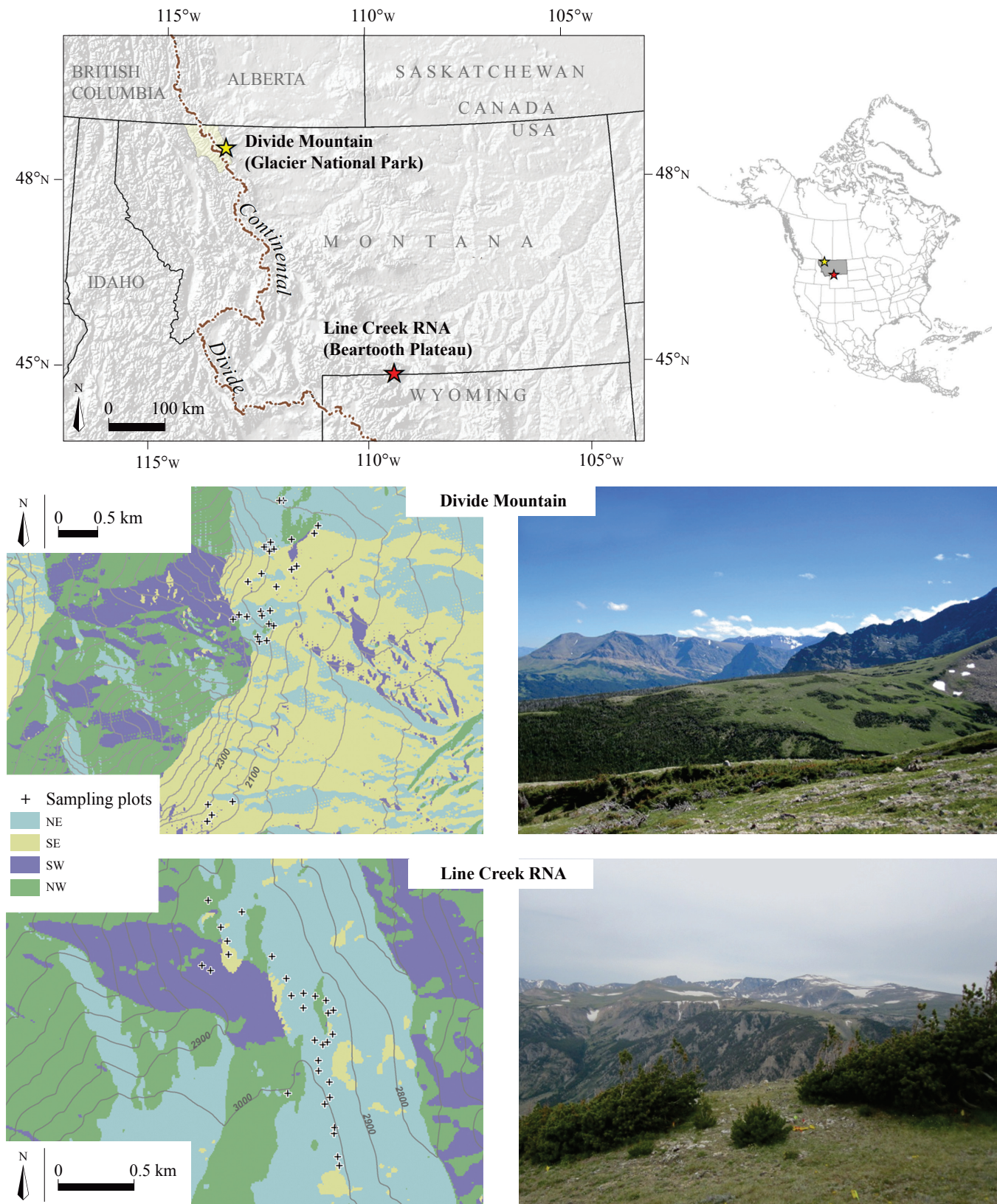


FIGURE 1. Location of 2 treeline study areas in the central and northern Rocky Mountains of Montana, USA. Detailed areas show sampling plot configuration in reference to the 4 aspect categories. Treeline top photo was taken at the base of Divide Mountain looking south, and treeline bottom photo in Line Creek Research Natural Area (RNA) looking north.

and surface boulders transported from cliff bases through active solifluction processes. Ongoing periglacial processes such as solifluction, needle ice activity, and rock sorting (Johnson & Billings, 1962; Butler & Malanson, 1989) have shaped the microtopography. Uneven microtopographic surfaces can facilitate conifer establishment and growth (Butler, Malanson & Resler, 2004; Resler, Butler & Malanson, 2005) through climatic amelioration. Furthermore, microtopographic features may serve as landmarks for the placement of *P. albicaulis* seed caches by Clark's nutcracker (Tomback, 1978). Each treeline community contained tree island colonies, groupings of sheltered conifers that established in the lee of more tolerant trees (such as *P. albicaulis*) or topographic features. Sampled tree species within the 2 treeline communities, coexisting in tree islands or growing solitarily in both study areas, include *P. albicaulis*, *A. lasiocarpa*, *Picea engelmannii* (Engelmann spruce), *Pinus contorta* (lodgepole pine), and *Pseudotsuga menziesii* (Douglas-fir).

Alternate host species for *Cronartium ribicola* (*Ribes* spp., *C. miniata*, *P. racemosa*) (McDonald & Hoff, 2001; McDonald *et al.*, 2006) are prevalent within both study areas. In particular, *Ribes* spp. plants were observed growing in and around our sampling plots, as well as within tree island patches (E. Smith-McKenna, unpubl. data). Undoubtedly, the spatial configuration of alternate host species in relation to disease manifestation in host trees is important; however, it is also complex and not fully understood at alpine treelines, and an exhaustive survey of alternate host species was precluded in this study due to level of sampling intensity, time, financial, and personnel limitations. Since Smith *et al.* (2011) previously found proximity to *Ribes* to be of lesser importance than topographic variables (e.g., surface curvature, aspect, and flow accumulation) in predicting disease canker intensity at the sampling plot level, we focus here specifically on abiotic controlling factors at the tree level.

SAMPLING CONFIGURATION

In July 2010, we sampled from thirty 15- × 15-m plots (following methods from Smith *et al.*, 2011) within the elevational limits of the ATE at each study area, for a total of 60 plots. The 225-m² plots captured a range of tree/tree island sizes and topographic variation within each treeline study area and maintained within-plot homogeneity in obvious environmental variation. Because slope orientation (e.g., aspect) has been linked to blister rust infection (Smith *et al.*, 2011), we stratified the sampling plots proportional to each treeline sampling area in 4 quadrants (northeast [0–90°], southeast [90–180°], southwest [180–270°], northwest [270–360°]). Using terrain and aerial imagery in a geographic information system (GIS), we delineated potential sampling boundaries at each treeline and calculated the areal proportion within each aspect quadrant. For example, plots were randomly dispersed across the Divide Mountain treeline study area on slopes facing NE (40%, 12 plots), SE (34%, 10 plots), SW (13%, 4 plots), and NW (13%, 4 plots). Within each treeline study area, we navigated with a topographic map and compass to determine aspect and used blind and random pin tosses

(toss distances >20 m) to designate sampling plot centres. We chose specific sample locations based on conifer presence, hiking accessibility, and safety (see Figure 1). Within each sampling plot, we 1) recorded *P. albicaulis* tree/tree island geospatial positions, 2) classified and quantified blister rust presence and canker intensity, and 3) characterized the microtopography by creating high-resolution digital elevation models, as detailed below.

PINUS ALBICAULIS AND BLISTER RUST CHARACTERIZATION

Within each sampling plot we inventoried all *P. albicaulis*, living and dead, and noted the characteristics of growth, size, and whether they grew solitarily or within tree islands. We inventoried dead *P. albicaulis* trees displaying 100% tree and canopy kill (dead needles), and noted whether blister rust caused the death (showing evidence of cankers) when *P. albicaulis* species could be identified. The majority of sampled *P. albicaulis* were krummholz trees less than 1 m in height, and we included all ages of trees (from seedlings to very old trees) in our inventories. Tree islands were defined as 2 or more trees growing contiguously, with spatially overlapping foliage. We considered a cluster of *P. albicaulis* stems apparently originating from a single nutcracker cache as a single tree (e.g., Resler & Tomback, 2008). We inspected each *P. albicaulis* tree for evidence of blister rust infection on the stems and branches and recorded the number and class of cankers, following Hoff (1992). Cankers were distinct from one another if separated by healthy tissue on the stem or branch (see Figure 2). We geolocated each *P. albicaulis* tree in our sampling plots in order to relate tree and disease characteristics (blister rust presence/absence and number of cankers) to the underlying microtopography surface variables, which were derived geospatially and are described in the following section.

Mountain pine beetles do not attack trees as small in stem diameter as krummholz *P. albicaulis* (Gibson *et al.*, 2008). However, larger diameter



FIGURE 2. Example of a) an inactive blister rust canker (cracked bark on upper stem, indicating past sporulation) and b) an active canker lower on the stem (aecial sacs have burst, releasing a bright orange dusting of spores). The third canker class, potential (the beginning stage of a canker, characterized by a swollen stem but no sporulation), is not shown (Hoff, 1992).

non-krummholz trees within sheltered areas of the lower ATE were attacked in our study areas, and particularly in the Line Creek RNA. None of our sampling plots contained these trees, and therefore infestation by mountain pine beetle was not a confounding factor in this study.

SPATIAL CHARACTERIZATION OF TOPOGRAPHIC VARIABLES

We characterized the microtopography of each sampling plot by recording elevation points across the plot's surface with a handheld Trimble GeoXT (Trimble Navigation Limited, Sunnyvale, CA, USA) global positioning system (GPS) unit in order to represent the varying surface gradients and topographic depressions. We differentially corrected all GPS points to submetre accuracy. Terrain complexity varied within each plot, as did the number of representative elevation points (on average, 600 points per plot) with which we created fine-resolution (on average, 0.08 m) digital elevation models (DEMs) following methods from Smith (2009) and Smith *et al.* (2011).

Submetre-resolution DEMs provide more detail in our 15-m sampling plots than the currently available USGS 10-m DEM, an important consideration since our scale of analysis is at the tree level, and many of our sampled trees are less than 1 m apart. Additionally, periglacial surface features are quite variable in the ATE (*e.g.*, Butler & Malanson, 1989; 1999), and fine-scale DEMs are important for characterizing the microtopographic variability (Allen & Walsh, 1996; Walsh *et al.*, 2003). For example, the size and type of fine-scale patterned ground topographic features characteristic of our sampling plots included about 14–18 cm turf-banked terrace risers (near Divide Mountain, *e.g.*, Butler, Malanson & Resler, 2004; Resler, 2004), about 17–35 cm frost hummocks, 1–3 m high solifluction terraces (near Line Creek RNA, *e.g.*, Johnson & Billings, 1962), and rock clasts ranging from 10 cm to very large boulders (Resler, 2004). In the severe ATE environment, this topographic variability creates microhabitats that ameliorate exposure to abiotic conditions and influence seedling survival, patchy vegetation, and snow retention patterns across the treeline landscape (Johnson & Billings, 1962; Oke, 1978; Holtmeier & Broll, 1992).

We derived all topographic variables using ArcGIS (v.9.3.1, ESRI, Redlands, California, USA) Spatial Analyst tools. From each of the 60 submetre DEMs, we calculated 4 submetre microtopographic raster surface variables influential to moisture and humidity conditions (slope, curvature, flow accumulation, and solar radiation) (ESRI, 2009). Topographic features that control water runoff and flow rate (*e.g.*, slope), cool air drainage, and soil moisture conditions are potentially influential for spore production and blister rust infection. Surface curvature indicates the reflection (convex, positive curvature) or retention (concave, negative curvature) of water. High flow accumulations indicate concentrated areas of drainage, particularly where water pools in depressions (Florinsky & Kuryakova, 2000). Solar radiation affects soil moisture conditions; surfaces with more sun exposure tend to have drier soils (which may be less favourable for blister rust).

The proximity of white pine trees to water features has been linked to blister rust infection (Van Arsdel, 1965;

White, Brown & Host, 2002; Smith *et al.*, 2011). Since alpine glaciers and snowmasses are present at the highest elevations within and surrounding both treeline study areas, and may provide moisture, we considered the distance to ice a potential blister rust correlate. Furthermore, we determined proximity distances from each *P. albicaulis* tree to the surrounding streams, lakes, and marsh and included these mesotopographic variables in our analysis. We derived these variables from high-resolution National Hydrography Dataset GIS layers, which are based on 1:24 000 USGS 7.5-min quadrangle maps (Simley & Carswell, 2009). Using the ArcGIS Spatial Analyst distance tool, we incorporated the terrain of a USGS 10-m DEM in the distance calculations to include the mountainous topography (mountains may influence wind-driven spores). We generated GIS raster map layers for the 4 mesotopographic distance variables (distance to streams, lakes, marsh, and ice).

Using a GIS raster-extraction tool, we extracted the 8 topographic values (4 micro- and 4 mesotopographic variables) for each geolocated *P. albicaulis* tree within each sampling plot in order to examine relationships between disease infection of trees and their corresponding topographic site characteristics. We used 8 independent topographic variables for the candidate model selection process: slope, curvature, flow accumulation, solar radiation, and distances to streams, marsh, lakes, and ice. Figure 3 shows an example of a sampling plot's variation in terrain and drainage relative to GPS locations of sampled *P. albicaulis* trees and their disease characteristics, and the detailed DEM microtopography variables used to extract values for each tree. Predictors for the dependent variables—blister rust presence (presence of cankers) and intensity (total number of cankers per infected tree)—were determined from the statistical analyses.

STATISTICAL ANALYSIS

We modeled blister rust presence and intensity as a function of topographic and environmental spatial variables for each sampled tree, within each sampling plot, for each study area. We used a zero-inflated Poisson (ZIP) regression multilevel model with random effects similar in model structure to that described by Hall (2000) and parameters estimated by Bayesian methodology. To run the statistical analyses, we coded a model simulation procedure similar in structure to that of Hadfield (2010) that used Markov Chain Monte Carlo (MCMC) methods and incorporated random effects, but implemented in MATLAB (version 7.9.0529 R2009b, The MathWorks Inc., Natick, Massachusetts, USA). It was necessary to apply ZIP regression methods since the dependent variable (number of cankers per tree) consisted of count data with many zeros (Lambert, 1992). The addition of a random-effects term accounted for correlated responses and sources of dependency, which can be problematic with multiple sources of variance (Hall, 2000), within and among the 30 sampling plots in each study area. By including random effects in our multilevel ZIP models, we were able to account for dependence of observations (blister rust cankers on each tree) within clusters (sampling plots). Multilevel random effects models can solve potential response dependency and multicollinearity simultaneously (Shieh & Fouladi, 2003).

Potential predictor variables were determined through stepwise selection procedures of 2 regression analyses: 1) a logistic regression with the presence *versus* absence of cankers as the dependent variable and 2) a linear regression with the number of cankers as the dependent variable, using SAS® software (version 9.2, SAS Institute Inc., Cary, North Carolina, USA). We screened the initial 8 variables for low Mallow's Cp, model fit statistics (AIC, SC, $-2 \text{ Log } L$), and multicollinearity (risk of multicollinearity was small: most variables displayed $r < 0.2$). Based on these statistical methods (Miller, 2002; Shieh & Fouladi, 2003; Muller & Fetterman, 2003), we selected 6 variables to include in the

ZIP models: curvature, solar radiation, distance to streams, distance to lakes, distance to marsh, and distance to ice (see Table I for variable ranges).

For each study area, we generated a ZIP model that uses 2 link functions, a logit (logistical) link function and a log (Poisson) link function, in order to model 2 processes at the same time. The models test 1) which variables are associated with the presence *versus* absence of blister rust (logistic or zero state) and 2) if blister rust is present, which variables predict blister rust intensity (Poisson state). The resulting ZIP regression model with random effects for the prediction of canker presence (p_{ij}) and mean intensity (λ_{ij}) is

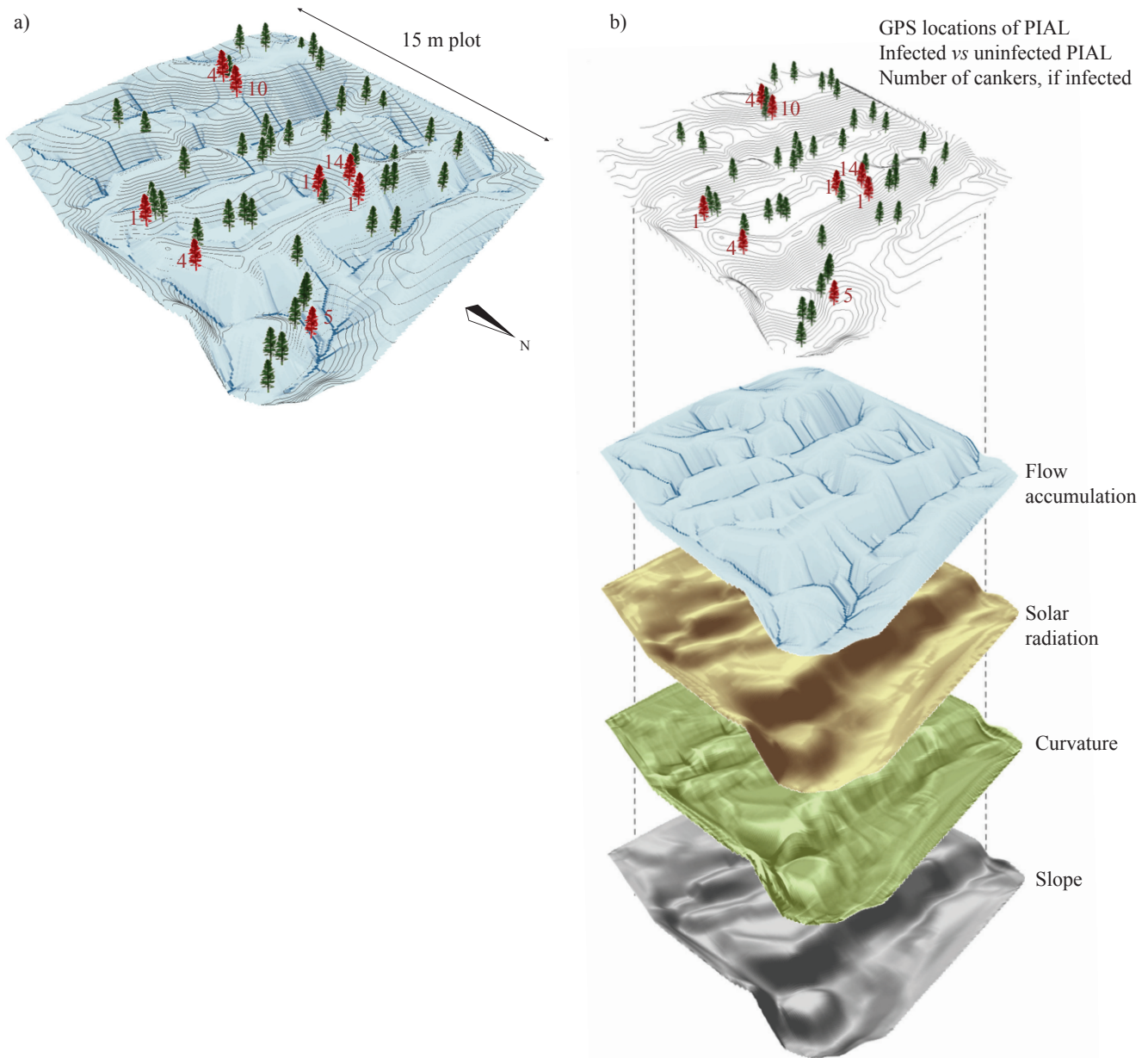


FIGURE 3. Examples of actual DEM microtopography variables and *P. albicaulis* (PIAL) tree spatial positions overlaid in a GIS. Maps are from sampling plot #15 in Glacier National Park. a) 3D view of the mapped terrain within a 15- × 15-m sampling plot, showing GPS positions of uninfected (green) and infected (red, noting canker count) trees relative to elevation contours and flow accumulation (drainage) concentrations on the underlying surface. b) Topographic raster values were extracted for each corresponding PIAL GPS location to correlate with blister rust infection presence and intensity (number of cankers) for each tree. Microtopography variables are shown here; mesotopographic variables (distance to water features) extend beyond view and are therefore not shown in this figure. The 3D maps were generated using ArcScene (v.10.1, ESRI, Redlands, California, USA).

expressed as follows (equations 1–3; for plot $i = 1, \dots, 30$; and for tree $j = 0, \dots, n_i$):

$$Y_{ij} \sim \begin{cases} 0, & \text{with probability } (1 - p_{ij}) \\ \text{Poisson } (\lambda_{ij}), & \text{with probability } p_{ij} \end{cases} \quad [1]$$

$$\log(\lambda_{ij}) = x'_{ij}\beta + w_i \quad [2]$$

$$\text{logit}(1 - p_{ij}) = z'_{ij}\gamma + u_i \quad [3]$$

Equation [1] defines Y_{ij} , which represents the number of blister rust cankers on the specific tree j within the specific sampling plot i . Within the log-linear (Equation [2]) and logistic (Equation [3]) regression equations, x'_{ij} and z'_{ij} represent the covariate vectors of the j^{th} tree within the i^{th} plot ($i = 1, 2, \dots, 30$; $j = 0, 1, \dots, n_i$). We estimated 2 separate covariance structures for canker intensity and presence in Equation [2] and [3], respectively. Parameters β and γ are the corresponding fixed-effects regression parameters. The vectors w_i and u_i are the subject random effects or random intercepts in 2 equations that are assumed to be independently and normally distributed since there is no spatial dependency between infected and uninfected trees (blister rust disease cannot spread from pine to pine [Mielke, 1943]). These equations are modified from Hall's (2000) ZIP random effects model, which incorporates random effects (a random intercept) only in the log (Poisson) link function.

To determine final models for each treeline study area, we implemented Bayesian procedures to produce model runs with credible interval ranges for each variable, assessed whether each interval range did not overlap with zero, and used non-informative priors (e.g., Daniels & Kass, 1999; Barnard, McCulloch & Meng, 2000). Model simulations consisted of 5000 iterations, with a burn-in period of 1000 (eliminating the first 1000 iterations), in order to reach model convergence. Bayesian Information Criterion (BIC) values indicate the strength of each variable's contribution to the model. We analyzed a series of candidate models for each study area by running one-variable ZIP models (Table II) and assessing BIC ranking in order to assist with the final model selection. One-variable ZIP models and BIC ranking are useful for providing descriptive analyses for exploratory purposes (e.g., Smith *et al.*, 2011).

The final ZIP regression random effects models define which topographic variables best explain the presence and intensity of blister rust infection at each treeline study area. The final models account for 1) a large number of zero values in the dependent variable, 2) any clustering between sampling plots, and 3) the number of *P. albicaulis* trees per sampling plot.

Finally, previous research has shown important differences in infection rates between solitary *P. albicaulis* and those growing within tree islands (Resler & Tomback, 2008). Therefore, we conducted a *G*-test of independence (Sokal & Rohlf, 1995) to compare infection rates against growth characteristics (solitary or tree island) at both Divide Mountain and Line Creek RNA.

TABLE I. Descriptive statistics for 6 potential variables input to each treeline model.

Divide Mountain (GNP)		
Variable	Range	Mean $\pm$ SD
Curvature ($1 \cdot 100 \cdot \text{m}^{-2}$)	−698.72–1 295.78	0.85 ± 161.99
Solar radiation ($\text{watt} \cdot \text{hour} \cdot \text{m}^{-2}$)	454 780–1 533 950	$1 140 277 \pm 221 029$
Distance to streams (m)	56–688	311 ± 166
Distance to ice (m)	8898–12 534	$11 422 \pm 801$
Distance to lakes (m)	1853–2781	$2 209 \pm 259$
Distance to marsh (m)	2469–4673	$3 532 \pm 559$
Line Creek RNA (Beartooth Plateau)		
Variable	Range	Mean $\pm$ SD
Curvature ($1 \cdot 100 \cdot \text{m}^{-2}$)	−1337.65–622.57	$−4.91 \pm 165.79$
Solar radiation ($\text{watt} \cdot \text{hour} \cdot \text{m}^{-2}$)	336 234–1 785 500	$1 358 323 \pm 271 740$
Distance to streams (m)	123–702	545 ± 128
Distance to ice (m)	974–3 077	1970 ± 530
Distance to lakes (m)	1792–2887	2430 ± 326
Distance to marsh (m)	3007–5107	4001 ± 528

TABLE II. Bayesian Information Criterion ranking of zero-inflated Poisson regression model variables for predicting blister rust cankers.

Divide Mountain (GNP)			
Independent variables	β^a	γ^a	BIC ^b criteria ranking (ranked lowest to highest)
Solar radiation	0.3165 ^c	−0.0040	1237.1
Distance to lakes	0.1557 ^c	−0.1881	1258.5
Distance to ice	−0.0925 ^c	−0.1442	1260.5
Distance to streams	0.0607	0.1572	1265.4
Curvature	0.0912	−0.0941	1265.9
Distance to marsh	0.0373	−0.0433	1268.9
Line Creek RNA (Beartooth Plateau)			
Independent variables	β	γ	BIC criteria ranking (ranked lowest to highest)
Distance to streams	−0.5162 ^c	−0.4683 ^c	957.9
Distance to ice	0.4556 ^c	0.4128 ^c	1017.4
Distance to marsh	0.4492 ^c	0.4200 ^c	1017.9
Distance to lakes	−0.4537 ^c	−0.3900 ^c	1019.3
Solar radiation	0.3644 ^c	0.3951 ^c	1061.4
Curvature	−0.1747 ^c	0.2887	1085.6

^a β , Poisson component parameter (test for canker intensity); γ , logistic component parameter (test for canker presence).

^b Bayesian Information Criterion (BIC). Variables ranked in order of importance, based on BIC. Model parameters with low BIC values are better predictors than parameters with high BIC values.

^c The parameter is significant at the 95% significant level. Parameters shown for the Poisson (β) and logistic (γ) components reflect the median values of the credible interval.

Results

PINUS ALBICAULIS AND BLISTER RUST SAMPLING RESULTS

We sampled a total of 913 *P. albicaulis* trees among the 60 plots (Table III). Divide Mountain had nearly double the number of sampled *P. albicaulis* trees (maximum per plot, 79) compared to

TABLE III. Tree and disease statistics for 2 treeline study areas, summarized by aspect category.

Divide Mountain (GNP)					
	Aspect				
	NE	SE	SW	NW	Total
Number of sampling plots	12	10	4	4	30
Number of sampled PIAL ^a (<i>N</i>)	293	111	49	132	585
Number of solitary PIAL	157	55	15	42	269
Number of tree island PIAL	136	56	34	90	316
% solitary PIAL	53.6%	49.5%	30.6%	31.8%	46.0%
% tree island PIAL	46.4%	50.5%	69.4%	68.2%	54.0%
Number of blister rust cankers found	256	160	26	63	505
Number of infected PIAL	64	38	9	27	138
% infected PIAL	21.8%	34.2%	18.4%	20.5%	23.6%
% infected solitary PIAL	17.2%	21.8%	13.3%	11.9%	17.1%
% infected tree island PIAL ^b	27.2%	46.4%	20.6%	24.4%	29.1%
Canker intensity (cankers per infected tree)	4.0	4.2	2.9	2.3	3.7
Canker intensity on solitary PIAL	2.0	2.6	1.0	1.8	2.1
Canker intensity on tree island PIAL	5.4	5.0	3.4	2.5	4.4
Line Creek RNA (Beartooth Plateau)					
	Aspect				
	NE	SE	SW	NW	Total
Number of sampling plots	14	8	4	4	30
Number of sampled PIAL ^a (<i>N</i>)	195	84	40	9	328
Number of solitary PIAL	103	63	18	7	191
Number of tree island PIAL	92	21	22	2	137
% solitary PIAL	52.8%	75.0%	45.0%	77.8%	58.2%
% tree island PIAL	47.2%	25.0%	55.0%	22.2%	41.8%
Number of blister rust cankers found	152	40	245	0	437
Number of infected PIAL	31	14	18	0	63
% infected PIAL	15.9%	16.7%	45.0%	0.0%	19.2%
% infected solitary PIAL	11.7%	9.5%	44.4%	0.0%	13.6%
% infected tree island PIAL ^b	20.7%	38.1%	45.5%	0.0%	27.0%
Canker intensity (cankers per infected tree)	4.9	2.9	13.6	0	6.9
Canker intensity on solitary PIAL	5.8	2.2	4.3	0	4.5
Canker intensity on tree island PIAL	4.4	3.4	21.1	0	8.7

^a PIAL: *Pinus albicaulis*^b *G*-test of independence (Sokal and Rohlf, 1995) shows blister rust infection is significantly higher for *P. albicaulis* growing in tree islands as opposed to solitary: Divide Mountain ($G = 11.85$, $df=1$, $P < 0.001$; Line Creek RNA ($G = 9.11$, $df=1$, $P < 0.01$).

Line Creek RNA (maximum per plot, 34) and more than double the number of infected trees. Divide Mountain showed a slightly higher percentage of blister rust-infected trees (23.6%) and lower canker intensity (maximum cankers per tree, 20) compared to Line Creek RNA (19.2%) (maximum cankers per tree, 62) (Table III). More *P. albicaulis* trees grew in tree islands than solitary at Divide Mountain; the reverse was true for Line Creek RNA. The *G*-test of independence indicated that, at both sites, *P. albicaulis* growing in tree islands had greater canker intensity and likelihood of blister rust infection than solitary *P. albicaulis* (Table III). Infection rate was highest on SE and SW aspects at Divide Mountain and Line Creek RNA, respectively (Table III).

SPATIAL CORRELATES OF BLISTER RUST PRESENCE AND INTENSITY

The final models showed significant results for both study areas (Table IV). Final model parameters were considered significant when the credible interval ranges (between lower to upper bounds) did not overlap zero. Solar radiation and distance to lakes were statistically significant in explaining some of the variance in blister rust canker intensity at both study areas. Proximal distance to ice was

an additional significant variable at Divide Mountain; at Line Creek RNA concave curvature and proximal distance to streams also explained variance in blister rust canker intensity (β , Poisson component, Table IV). At Line Creek RNA, long distances to ice explained a portion of variance in blister rust presence (γ , logistic component) on *P. albicaulis* trees. Estimated parameter values for distance to lakes at Divide Mountain and curvature at Line Creek RNA (Table IV) indicated that these variables were linked to the presence of blister rust cankers. However, results in the one-variable zero-inflated Poisson ZIP models were not significant for either variable (Table II). Therefore, we considered the relationships to be weak and likely due to interaction with other variables (e.g., Lambert, 1992).

Discussion

Topography is important at alpine treelines in creating varied microclimates crucial for conifer establishment, survival, and colonization (e.g., Butler, Malanson & Resler, 2004; Resler, Butler & Malanson, 2005), and there appears to be a link between fungal infection and topographic features that ameliorate the harsh conditions in the ATE. These conditions may potentially

TABLE IV. Results of final zero-inflated Poisson random effects models for Divide Mountain and Line Creek RNA treeline study areas. Significant parameter values for the prediction of blister rust cankers are shown.

Divide Mountain Model							
Variable	β^a			γ^a			BIC ^c rank
	Lower ^b (2.5%)	Median (50%)	Upper (97.5%)	Lower ^b (2.5%)	Median (50%)	Upper (97.5%)	
Intercept	1.0154	1.1313	1.2440	-1.3411	-1.1284	-0.9226	Most important ↓ Least important
Solar radiation	0.1870	0.3002	0.4087	—	—	—	
Distance to lakes	0.1921	0.3218	0.4356	-0.4367	-0.2178	-0.0148	
Distance to ice	-0.2435	-0.1632	-0.0782	—	—	—	
Distance to streams	—	—	—	—	—	—	
Curvature	—	—	—	—	—	—	
Distance to marsh	—	—	—	—	—	—	
Line Creek RNA Model							
Variable	β			γ			BIC rank
	Lower (2.5%)	Median (50%)	Upper (97.5%)	Lower (2.5%)	Median (50%)	Upper (97.5%)	
Intercept	1.0273	1.2198	1.3855	-1.7145	-1.3750	-1.0623	Most important ↓ Least important
Distance to streams	-0.5016	-0.3966	-0.3024	—	—	—	
Distance to ice	—	—	—	0.1373	0.4305	0.7390	
Distance to marsh	—	—	—	—	—	—	
Distance to lakes	-0.6800	-0.5162	-0.3587	—	—	—	
Solar radiation	0.1256	0.2439	0.3727	—	—	—	
Curvature	-0.9244	-0.7103	-0.4938	0.2459	0.6931	1.2101	

^a β , Poisson component parameter (test for canker intensity); γ , logistic component parameter (test for canker presence).

^b Credible interval (2.5%–97.5%) for β and γ . Significant parameter values are shown for credible interval ranges that do not overlap zero.

^c Best BIC results determined the final treeline models, variables are ranked in order of importance. Divide Mountain final model BIC = 1221.8; Line Creek RNA final model BIC = 907.1.

influence (by inhibiting and/or promoting) the distribution and structure of some Rocky Mountain treeline *P. albicaulis* communities (e.g., Resler & Tomback, 2008; Smith *et al.*, 2011). Comparing our 2 *P. albicaulis* treeline study areas, we found that the percentage of blister rust–infected *P. albicaulis* varied slightly, but canker intensity differed considerably. Furthermore, between Divide Mountain and Line Creek RNA, there were both geographic differences and similarities in topographic variables that influence blister rust infection.

GEOGRAPHIC VARIATION OF BLISTER RUST IN TREELINE *P. ALBICAULIS*

Our results are consistent with findings from other studies that have examined the influence of landscape factors on blister rust infection in white pine species in locations with comparatively cold and dry climates (e.g., White, Brown & Host, 2002; Smith *et al.*, 2011). We found that SE aspects at Divide Mountain and SW aspects at Line Creek RNA showed the highest blister rust infection levels at alpine treeline. Though Larson (2011) found the highest blister rust infection levels in subalpine *P. albicaulis* stands on north-trending slopes, he sampled west of the Continental Divide, where climates are generally wetter and milder than east of the Divide (Finklin, 1986). Blister rust incidence could also be inhibited by cold conditions at key times of development or spore dispersal (i.e., shorter growing seasons and fewer frost-free days [e.g., Campbell & Antos, 2000] on northern slopes: White, Brown & Host, 2002)—a relationship that could hold true in extremely cold and dry

high-elevation environments east of the Continental Divide (Finklin, 1986; Walsh, Malanson & Butler, 1992) and in our treeline study areas. These findings illustrate the importance of geography in blister rust dynamics.

The spread of blister rust to cold, dry treeline communities was originally considered unlikely (e.g., Campbell & Antos, 2000) because certain stages of the blister rust life cycle require both moderate temperatures and high humidity (e.g., McDonald & Hoff, 2001; Geils, Hummer & Hunt, 2010). However, in this study nearly one-fourth of *P. albicaulis* showed signs of blister rust infection. Infection rates at Divide Mountain were slightly higher than those at Line Creek RNA, which may be explained by variation in treeline-specific factors, such as microclimate and tree island density, which may relate to microtopography. For example, weather stations placed at our treeline study areas (July–September 2010) recorded more frequent readings and longer durations of high relative humidity (>90%) at Divide Mountain compared to Line Creek RNA (Franklin, 2011). Average and maximum wind speeds and gusts measured at unprotected microsites at Divide Mountain during the summers of 2011 and 2012 exceeded measurements at Line Creek RNA (J. Pyatt, unpubl. data). Furthermore, monthly summer climate averages for the last 30 y (June–September, 1980–2010) showed wetter conditions and milder temperatures at Divide Mountain (precipitation = 76.3 mm, $T_{\text{mean}} = 11.2$ °C, $T_{\text{dewpoint}} = 0.02$ °C) than Line Creek RNA (precipitation = 44.5 mm, $T_{\text{mean}} = 9.6$ °C, $T_{\text{dewpoint}} = -3.2$ °C) (PRISM Climate Group, 2011).

Additionally, tree island density (including conifer species other than *P. albicaulis*) and *P. albicaulis* density for the same plots as measured in this study were consistently higher at Divide Mountain than Line Creek RNA (Franklin, 2011).

However, when comparing our results for Divide Mountain to other reported infection rates solely within the Northern Divide Ecosystem, we found the infection rate to be comparatively low for both regional treeline and subalpine stands. For example, we found a 24% infection rate for treeline at Divide Mountain as compared to 73–85% overall for subalpine stands in the Northern Divide Ecosystem (Kendall & Keane, 2001; Smith *et al.*, 2008). Furthermore, Smith *et al.* (2011) found that infection rates in stands near White Calf Mountain (adjacent to Divide Mountain) were the second lowest among 6 treeline sites they sampled in Glacier National Park. The comparably low treeline infection rate at Divide Mountain is interesting considering that subalpine *P. albicaulis* communities of the Northern Divide Ecosystem consistently have the highest rate of blister rust infection yet found across the range of *P. albicaulis*. Further research is warranted to investigate why Divide Mountain has relatively low infection levels within a region with very high spore loads, but local factors such as wind direction and complex wind patterns (in addition to wind speed) and geographic variation in genetically resistant trees may contribute.

Blister rust arrived somewhat later (in 1944, Kendall & Asebrook, 1998) in the GYE as compared to the Northern Divide Ecosystem (in 1938), as reported by Mielke (1943). Infection levels in the GYE remained low (0–1%) (Carlson, 1978) until the mid-1990s, when infection levels began to rapidly increase (Jean *et al.*, 2011; Bockino & Tinker, 2012). Evidence exists for rapid intensification of blister rust infection in the GYE in the last 20 y, with 2–13% infection rates reported in the mid-1990s (Kendall & Keane, 2001) and an overall 20% infection rate reported by 2007 (Jean *et al.*, 2011).

Our overall blister rust infection rates for treeline in the Line Creek RNA (19.2%) compare well with the overall current infection rate for the GYE region (about 20%). One possible explanation for the increase in current infection rates may be attributed to increased efforts in blister rust surveys, including surveys within GYE forests that had remained inaccessible in years past (*e.g.*, Hatala *et al.*, 2011). The increase in blister rust in the GYE might also be attributed, at least in part, to climatic warming trends reported for western Montana (Pederson *et al.*, 2010). Topography that creates conditions amenable to blister rust infection, combined with continued warming trends, could create favourable conditions for blister rust spore production and transmission, even at treeline.

We found that infection intensity is higher in tree island *P. albicaulis* than solitary *P. albicaulis* (Table III), which is similar to the findings in our previous research studies (*e.g.*, Resler & Tomback, 2008; Smith *et al.*, 2011). Microclimates within tree islands (*e.g.*, higher humidity) may favour blister rust spore development and infection. Van Arsdel (1965; 1972) noted higher blister rust infection of *P. Strobus* (eastern white pines) trees within forest stand interiors, as opposed to edges, attributing higher blister rust spore development and infection transmission to cool and

moist air currents within the stand. At treeline, the spatial clustering of trees within tree islands could similarly facilitate blister rust infection by increasing humidity, reducing ultraviolet light and wind exposure, sheltering alternate host plants, or creating local eddies that could circulate spores for infection transmission. Larger patch sizes (dimensions) of *P. albicaulis* tree islands have also been correlated with higher blister rust infection (Resler & Tomback, 2008; Franklin, 2011; Smith *et al.*, 2011), and tree islands might provide a larger surface area for intercepting wind-borne *Cronartium ribicola* spores. These conditions and characteristics are also a function of the microtopography underlying these treeline communities, in terms of modifying abiotic conditions for growth and influencing tree island spatial pattern and structure. In our study areas it appears that these conditions, characteristics, and the coexistence of host and alternate hosts act in synergy to support blister rust disease in an otherwise harsh environment.

INFLUENTIAL TOPOGRAPHIC FACTORS

The topographic factors influencing blister rust presence and intensity on *P. albicaulis* differed between the treeline study areas (see final ZIP models, Table IV), which highlights the importance of meso- and microtopography factors, in conjunction with climate, in determining blister rust infection. The Divide Mountain model suggests that if blister rust is present, *P. albicaulis* trees growing on surfaces with higher solar radiation exposure and that are farther from lakes but proximal to ice are likely to be infected with more cankers per tree than those in areas without these characteristics. Solar radiation aids the development of blister rust in the cold climate of our treeline study areas. Although glaciers are quite far from our Divide Mountain plots (Table I), and lakes are closer, the signs of their influence are opposite (although both are hypothesized sources for moist air). These variables are probably surrogates for the broader influence of topography (and glaciers and lakes themselves) on wind. Wind is a major factor affecting both snow and tree distribution on exposed slopes in the ATE (Arno & Hammerly, 1984; Malanson *et al.*, 2009), as well as the transportation of blister rust spores (Frank *et al.*, 2008). Van Arsdel (1965) attributed higher blister rust intensity in *P. Strobus* trees located on high ridges further from lakes to the role of wind for spore dispersal, which could also be true at Divide Mountain. In fact, the Divide Mountain plots are located about 7 km from and at an elevation about 900 m higher than the very large Saint Mary Lake (15.9 km² in area), emphasizing the importance of topographic position for the potential interception of wind-dispersed blister rust spores.

The final model for Line Creek RNA uncovered more significant relationships between blister rust presence and intensity and topographic variables than did the Divide Mountain model. The Line Creek RNA model suggests that trees farther from glaciers are more likely to be infected with blister rust, and that infected *P. albicaulis* trees tend to have more cankers in areas closer to streams and lakes, with higher solar radiation, and with concave surface curvature. Although glaciers and ice masses are closer to *P. albicaulis* trees at Line Creek RNA (Table I) compared to Divide Mountain, their influence probably has a surrogate explanation similar to that of the Divide Mountain model.

The high and exposed topographic position of the Line Creek RNA treeline community, a position potentially favourable for the interception of wind-dispersed blister rust spores, may compensate for the long distances to moisture sources. The relationship between canker intensity and the variables concavity and proximity to streams and lakes may indicate the combined importance of microtopography and surrounding moisture sources. Together, these features potentially provide the humidity and alternate host plant habitat necessary for blister rust transmission and infection in the Line Creek RNA study area, an area with more arid summers than Divide Mountain (PRISM Climate Group, 2011).

Other researchers have found similar trends between fungal infection and moisture-associated topographic variables at varying scales. For example, Kearns and Jacobi (2007) examined stand-level relationships between blister rust and *P. flexilis* and found that plots located in concave areas where moisture or cold air collects (*i.e.*, valley bottoms and foot slope positions) had higher infection rates than other topographic positions. Wilds (1997) found that fine-scale concave topography and proximity to streams were significant moisture characteristics related to the severity of dogwood anthracnose fungal disease (caused by the fungus *Discula destructiva*). In addition to moist conditions, concave microtopography potentially provides protective seedling microsites for *P. albicaulis* regeneration in the ATE. At a finer scale, the tendency of *P. albicaulis* to grow in protected microsites such as topographic depressions or leeward of rocks (Resler, Butler & Malanson, 2005) may also increase susceptibility to blister rust infection. Such microsites favouring seedling regeneration may also hinder *P. albicaulis* by providing moist conditions that increase the risk of fungal infections (Mellmann-Brown, 2005) such as blister rust, because depressions may also harbour abiotic conditions ideal for blister rust spore development. These combined conditions and processes may create “the perfect storm” for supporting the disease at treeline.

Conclusion

One major challenge for environmental change research is to understand how future vegetation changes will evolve in mountain ecosystems, where physiographic complexity creates highly variable environmental conditions over fine spatial scales (Shafer, Bartlein & Whitlock, 2005). The dynamics of pathogens, coupled with climate change, is likely to further complicate understandings of such vegetation change, particularly where keystone species are affected (Tomback & Resler, 2007). Our model results provide some insight (at a fine scale) as to where treeline patterns may be disrupted first by blister rust, in addition to climate change impacts. Between our 2 study areas, we found both differences and similarities in topographic and moisture-associated factors that influence blister rust infection of *P. albicaulis* trees, and so a landscape approach designed to uncover factors affecting forest disease (*e.g.*, Holdenrieder *et al.*, 2004) set in a geographic context with multiscale analyses is appropriate. The establishment of permanent plots at alpine treelines would facilitate understanding of the long-term consequences of blister rust on

tree island development, in the context of treeline dynamics, particularly if combined with remote sensing and climate models in mountain areas. Identifying treelines with potential blister rust severity could aid restorative management applications (some of which are ongoing, *e.g.*, Keane *et al.*, 2012), and might also aid proposed assisted migration (*e.g.*, McLane & Aitken, 2012). For example, blister rust-resistant *P. albicaulis* seedlings could be planted to restore treeline ecosystems, as they are being used at subalpine elevations (*e.g.*, Tomback & Achuff, 2010). Targeted efforts could mitigate the potential combined impact of blister rust and climate change on the treeline ecotone.

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